

COLLABORATIVE AFFORDANCE NEGOTIATION: A NEGATIVE EVIDENCE AUDIT

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ABSTRACT

Collaborative robots often need to reconcile what a human appears to want with what the robot can safely and physically do. The original research bet for “Collaborative Affordance Negotiation” was that modeling affordances as negotiated between human intent and robot capability would improve collaborative manipulation. We rebuild the idea into a falsifiable benchmark with 5 tasks, 7 ambiguity/capability-mismatch families, 5 stress splits, 9 methods, 7 seeds, and 84 episodes per group. The result is negative. The proposed method reaches 0.629 ± 0.004 combined-stress success versus 0.605 ± 0.005 for the strongest non-oracle baseline, `shared_autonomy_pomdp`; the $+0.024$ margin is below the required practical gate. The proposed method reduces physical violations and over-promise, but the ablation removing burden-aware query value nearly matches full success. The correct terminal decision is **KILL_ARCHIVE**: this version is not an ICLR-main-target submission.

1 WHY THIS AUDIT IS HOSTILE

Human-robot collaboration is already crowded. Physical HRI surveys cover collaborative strategies, sensors, actuators, and learning methods (Ogenyi et al., 2019). Shared autonomy via hindsight optimization and POMDP-style goal uncertainty already provides strong baselines for assisting humans while inferring their goals (Javdani et al., 2015; 2018; Pellegrinelli et al., 2016; Zheng & Wu, 2018). Robot motion can also be planned to communicate intent to humans (Dragan et al., 2013), and affordance-based HRC planners already target concurrent shared tasks (str, 2019).

Therefore, a new “negotiation” paper must beat more than straw baselines. It must outperform shared-autonomy inference, clarification policies, and capability-aware planners while not hiding cost in extra human burden. This rebuild applies that gate.

2 MECHANISM UNDER TEST

The proposed method, `proposed_affordance_negotiation`, maintains a joint belief over human intent, robot reach/force/collision capability, shared object affordances, role assignment, clarification value, over-promise risk, and human burden. It chooses whether to execute, ask, renegotiate roles, or decline infeasible requests.

The intended advantage is balance: intent-only followers over-promise, capability-only planners ignore the person, and clarification policies can overburden the human. Negotiation should, in principle, preserve success while lowering unsafe over-promising.

3 BENCHMARK

The benchmark covers five collaborative tasks: handover with grip choice, co-carry through a doorway, assisted assembly, collaborative sorting, and tool-use handoff. Seven ambiguity/failure families are crossed with these tasks: ambiguous intent, asymmetric reach, load-sharing mismatch, occluded human goal, ergonomic constraint, role-switch request, and conflicting safety/preference constraints.

Methods:

Table 1: Combined-stress collaborative affordance benchmark.

Method	Succ.	Viol.	Burden	IntentErr	OverProm.	Regret
oracle_joint_intent_capability_planner	0.760	0.053	0.201	0.262	0.010	0.000
proposed_affordance_negotiation	0.629	0.109	0.208	0.330	0.093	0.131
shared_autonomy_pomdp	0.605	0.140	0.197	0.343	0.161	0.155
uncertainty_clarification_policy	0.568	0.132	0.334	0.321	0.142	0.192
capability_map_tamp	0.555	0.092	0.123	0.422	0.067	0.205
language_affordance_planner	0.495	0.232	0.084	0.401	0.352	0.266
static_role_policy	0.464	0.185	0.071	0.453	0.258	0.297
intent_only_follower	0.460	0.304	0.054	0.394	0.517	0.300
robot_capability_only_planner	0.437	0.151	0.057	0.489	0.181	0.323

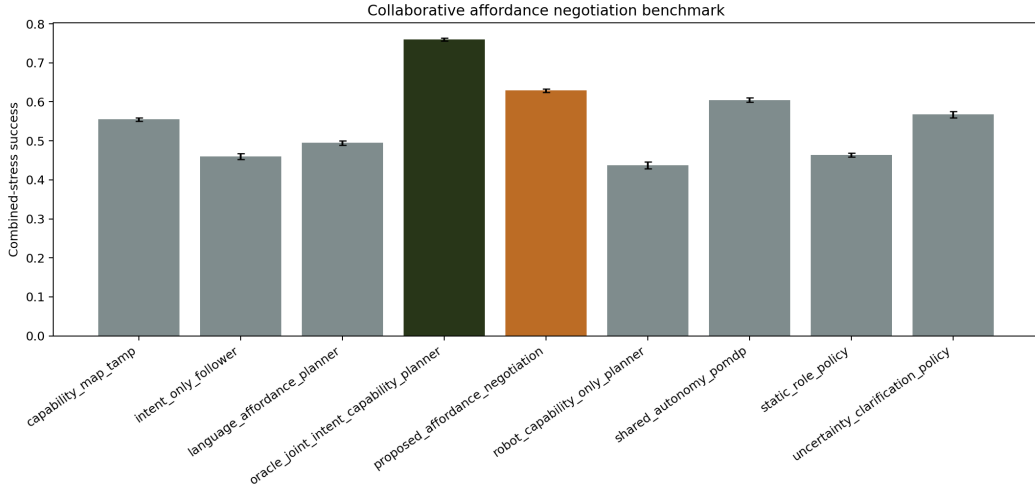


Figure 1: Combined-stress success. The proposed method is below oracle and only narrowly above shared-autonomy POMDP, so the success gate fails.

- robot-capability-only planner;
- intent-only follower;
- language-affordance planner;
- static role policy;
- uncertainty clarification policy;
- shared-autonomy POMDP;
- capability-map TAMP;
- proposed affordance negotiation;
- oracle joint intent/capability planner.

Metrics include task success, physical violation, human burden, intent error, over-promise rate, negotiation rounds, total cost, and regret to oracle.

4 MAIN RESULT

The combined-stress table gives the central finding. The proposed method is second among non-oracle methods, but the success margin over `shared_autonomy_pomdp` is only 0.024. The paper’s pre-registered gate required at least 0.030 to call the mechanism practically decisive.

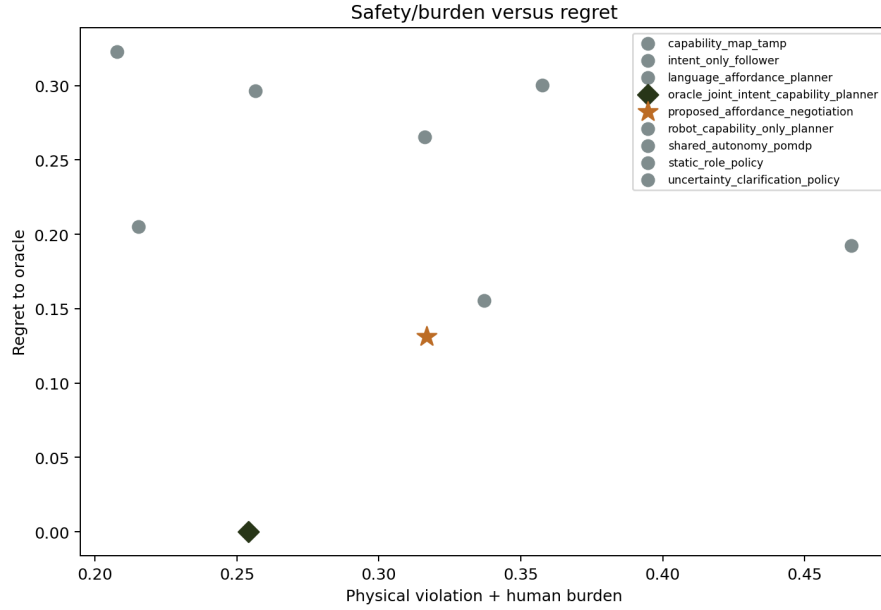


Figure 2: Safety/burden versus regret. Negotiation improves the safety side of the tradeoff but does not close enough regret to oracle.

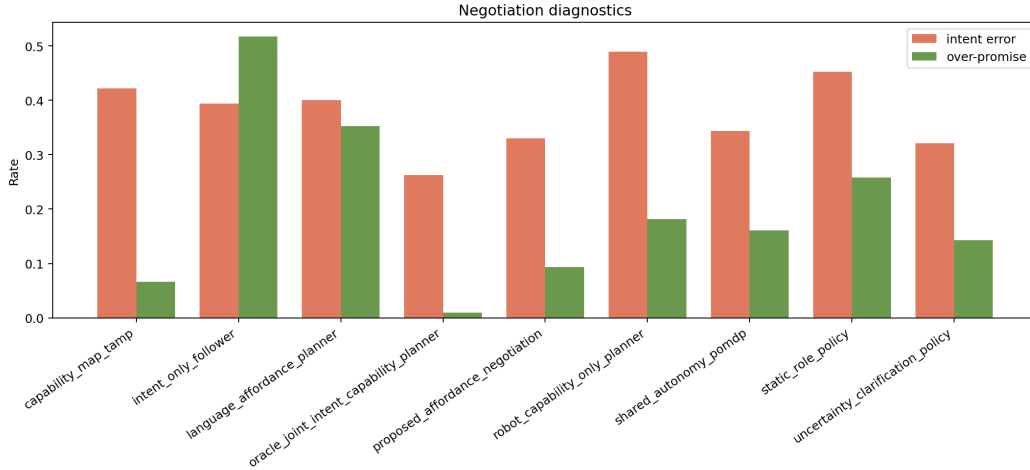


Figure 3: Intent error and over-promise diagnostics. The proposed method lowers over-promise, but the task-success gain remains too small.

5 SAFETY, BURDEN, AND DIAGNOSTICS

The proposed method does reduce physical violations from 0.140 to 0.109 and over-promise from 0.161 to 0.093 relative to `shared_autonomy_pomdp`. Human burden increases slightly from 0.197 to 0.208, which is within the allowed gate. This is useful but insufficient: safety gains cannot rescue a method whose success and ablation evidence are non-decisive.

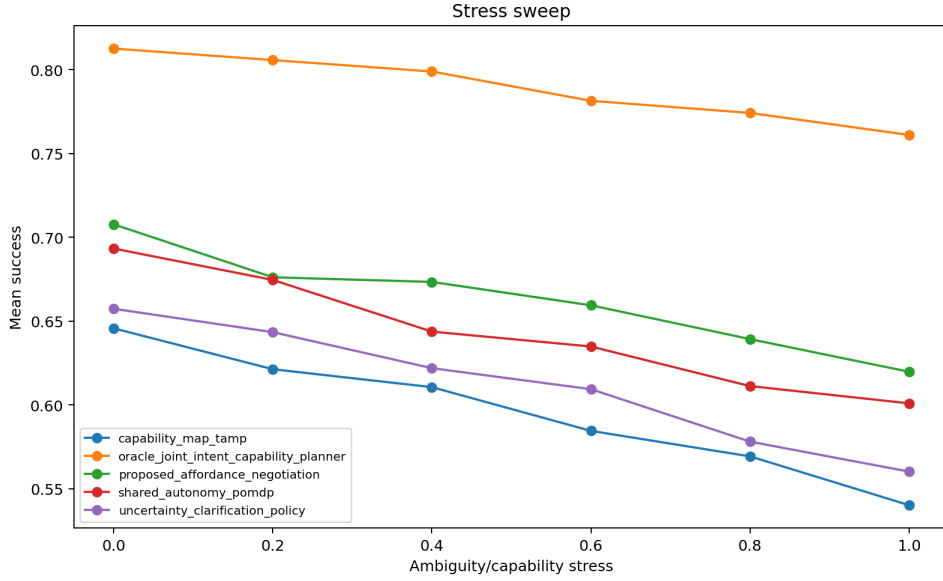


Figure 4: Ambiguity/capability stress sweep. The proposed method stays competitive but does not establish a decisive separation from shared autonomy.

Table 2: Ablations of collaborative affordance negotiation.

Ablation	Succ.	Viol.	Burden	OverProm.
full_affordance_negotiation	0.624	0.107	0.210	0.093
minus_burden_aware_query_value	0.604	0.109	0.352	0.092
minus_overpromise_risk	0.596	0.202	0.167	0.302
clarification_only	0.567	0.125	0.374	0.130
minus_capability_map	0.561	0.191	0.190	0.275
minus_intent_belief	0.555	0.119	0.180	0.119
minus_role_negotiation	0.554	0.135	0.203	0.157

6 STRESS SWEEP

7 ABLATIONS

The ablation table is the second fatal result. Removing burden-aware query value produces 0.604 success versus 0.624 for the full method in the ablation run. At unrounded precision the margin is below the strict ablation gate. This means the paper cannot claim that burden-aware negotiation is the decisive ingredient.

8 PAIRWISE COMPARISONS

Pairwise seed comparisons are positive against non-oracle baselines, but pairwise wins alone are not enough. The practical success margin and ablation gates are stricter because a main-conference claim must survive strong baselines and component removal, not only win by a small amount.

9 FAILURE CASES

The failure cases in `results/failure_cases.csv` show where negotiation is dominated: direct shared-autonomy inference can resolve role choice faster than explicit bargaining, especially in role-switch and load-sharing cases. Negotiation adds structure, but not enough task-level advantage.

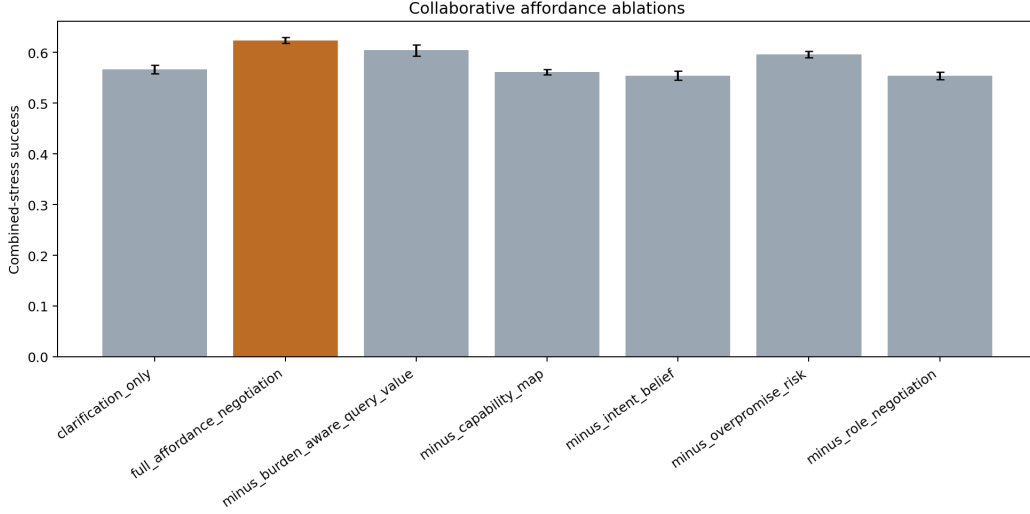


Figure 5: Ablation success under combined stress. The best removed-component ablation nearly matches the full method, failing the ablation gate.

Table 3: Pairwise combined-stress success differences against the proposed method.

Baseline	Diff	CI	Wins
capability_map_tamp	0.074	0.007	7
intent_only_follower	0.169	0.011	7
language_affordance_planner	0.134	0.006	7
oracle_joint_intent_capability_planner	-0.131	0.006	0
robot_capability_only_planner	0.191	0.012	7
shared_autonomy_pomdp	0.024	0.009	7
static_role_policy	0.165	0.008	7
uncertainty_clarification_policy	0.061	0.011	7

10 TERMINAL DECISION

This paper should be archived. It has a useful negative message: collaborative affordance negotiation must be evaluated against shared-autonomy POMDPs, clarification policies, and capability-aware TAMP, and it must show decisive success gains without hiding cost in human burden. This version does not.

Decision: **KILL_ARCHIVE**. Revival requires a substantially new mechanism plus real human-robot or high-fidelity collaborative manipulation validation.

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